

Lecture 1

Introduction to Machine Learning

Welcome to CS189/289

EECS 189/289, Fall 2025 @ UC Berkeley
Joseph E. Gonzalez and Narges Norouzi

Two Things...



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Pre-Semester Survey

Take a minute to fill out the [pre-semester survey](#) now if you haven't already!

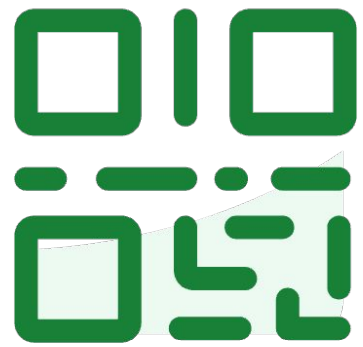


Spotify Playlist

We created [this collaborative Spotify playlist](#) to be played before class. Share your favorite songs with the class!



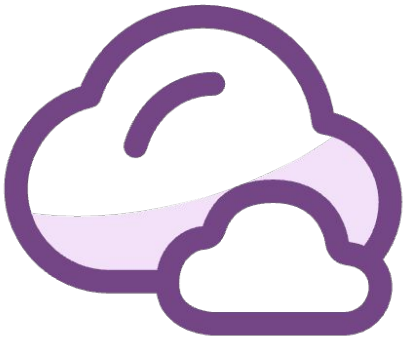
Do not edit
How to change the
design



**Join at slido.com
#2630936**

① The Slido app must be installed on every computer you're presenting from

slido



What emoji best describes your mood today?

Roadmap

- Introductions
- CS 189/289 Goals and Plans
- What Is Machine Learning?
 - Definition
 - History
- The ML Process (Lifecycle)
- Logistics



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Introductions

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Narges Norouzi



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- **Associate Teaching Professor** at EECS UC Berkeley (2022-now).
 - A member of [BAIR](#) and [ACE](#).
- **Associate Dean** in the College of Computing Data Science, and Society!
- **Research:** ML and ML tools in education
 - Askademia (www.askademia.org): Real-time video question-answering
 - [Edison](#)
 - Personalized tutoring with Knowledge Tracing capability
 - [61A bot](#)
 - [INSIGHT](#)
- Also taught ML, AI, Deep Learning + Data 100!
- This semester, I'm:
 - Teaching this course, advising my students, and leading EECS AI Task Force.
 - Doing my Associate Dean job!!!



[Pronunciation
of my name](#)

Joey Gonzalez



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- **Associate Research Professor** at EECS UC Berkeley (2016-now).
 - Co-directory of the [Sky Computing Lab](#) and a member of [BAIR](#)
- **Research:** AI and AI-Systems
 - Training and Inference Systems (e.g., [Clipper](#), [vLLM](#), [Alpa](#))
 - Developing new LLMs and Agents (e.g., [Gorilla LLM](#), [MemGPT](#), and [Vicuña](#))
 - Evaluating LLMs (e.g., [Chatbot Arena](#) and [LLM-as-a-judge](#))
 - Visual Language Reasoning and Data Analysis ([Lisa's](#) many Projects)
 - Robotics and Autonomous Driving Systems ([ERDOS](#))
 - Large-Scale Data Processing Systems (e.g., [Apache Spark](#), [GraphLab](#), [Ray](#))
- **I also teach:** Data-8, Data-100, and CS289-162 (Graduate AI Systems)
- This semester, I'm:
 - Teaching this course, running the Sky Lab and launching a new GenAI systems lab.
 - Vice Chair Grad Matters, UCB Division Council, and Chair campus GenAI working group
 - Advising a bunch of AI companies *and* coaching my daughter's 2nd grade robotics team



CS 189/289 Goals and Plans



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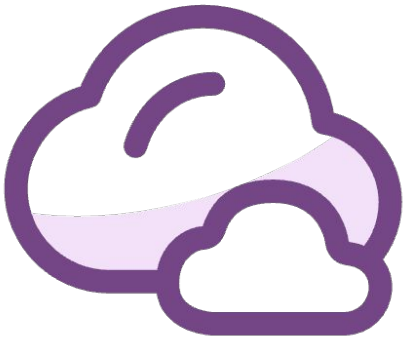
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Why Are We Excited to Teach CS189?

- **The field shifted under our feet** → we're redesigning now so we surf, not chase.
- **We listened to student pain points** → we're tightening the storyline to **math** → **code** → **experiments**, with unified notation and right-sized and reasonable prerequisites.
- **Classroom work should feed research** → projects will connect to live problems so **students** → **collaborators**.
- **Berkeley should set the bar** → we aim to produce a shareable resource that others can adopt and adapt.
- We are trying **MANY NEW THINGS** some things will break... but we hope you will enjoy this class.



Why are you here? What do you want to get from this class?



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By the End of CS189, You Should:

- Have a **rigorous foundation** in core ML concepts and algorithms, connecting the math to the methods we use.
- Be able to **implement, train, and debug** standard models in Python/PyTorch and move from data prep to experiments.
- Be able to **design sound evaluations** and document your work so results are **reproducible and responsible**.
- Be **prepared for advanced courses and research**.

These are ambitious goals!

Accel in Advanced ML Courses



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Advanced ML course:

- Computer vision (180/280), NLP (288), RL (285), Statistical Learning Theory (281), and Many Special Topics Courses on AI (194/294)

Require understanding of basic concepts:

- Model design
- Stochastic Gradient Descent
- Maximum Likelihood
- Loss Structure
- Supervised Learning
- Unsupervised Learning
- Classification
- Regression
- Regularization and the Bias/Variance Tradeoff
- Dimensionality Reduction

What Is Machine Learning?

- Introductions
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What Is Machine Learning (ML)?

Software systems that improve (learn) through data.

Classic Example: What is Spam?

- Difficult to define
- Easy to demonstrate

Classic Example: Face detection?

- Difficult to program
- Easy to demonstrate

Basic Recipe:

- Use examples (**data**) to teach (**fit**) a **model**.
- Use the **model** to **make decisions**.

What Is Artificial Intelligence?

Artificial intelligence (AI) refers to the capability of **computational systems to perform tasks typically associated with human intelligence**, such as learning, reasoning, problem-solving, perception, and decision-making.

– Wikipedia (2025)

The primary technology behind modern AI is **Machine Learning (ML)**

Artificial Intelligence Is the **Goal** Machine Learning Is the **Method**

Terms are often used interchangeably (even by experts).

An entrepreneur's note on AI marketing

If you are selling

- ❑ **Machine Learning** then you call it **Deep Learning**. *(2015-2020)*
- ❑ **Deep Learning** then you call it **AI**. *(2020-2024)*
- ❑ **AI** you call it an **Agent**. *(2024-Now)*



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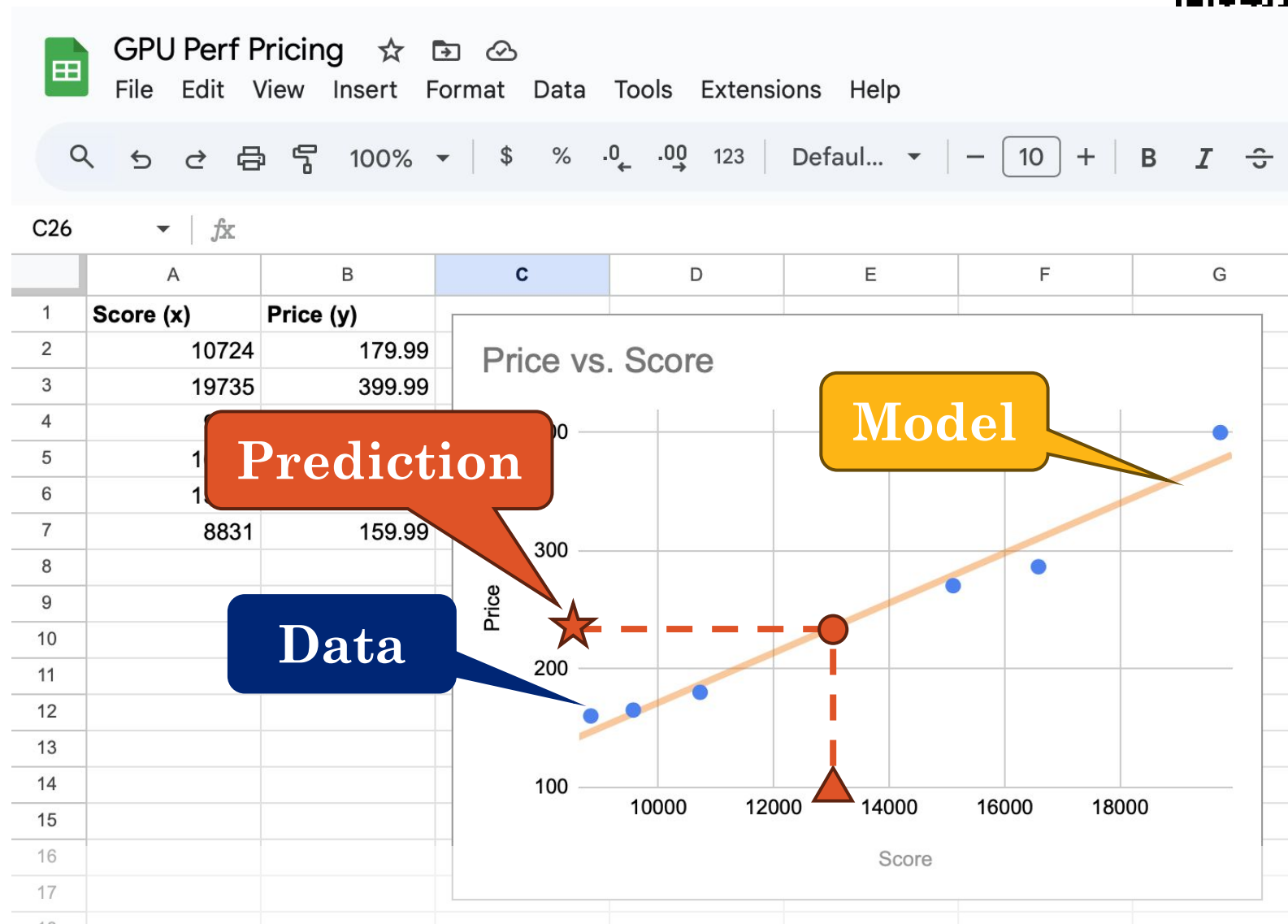
You have (probably) already
done **Machine Learning**

Linear Regression is Machine Learning

Basic Recipe:

Training: Use **data** to teach (fit) a **model**.

Inference: Use the **model** to make **predictions** (decisions).



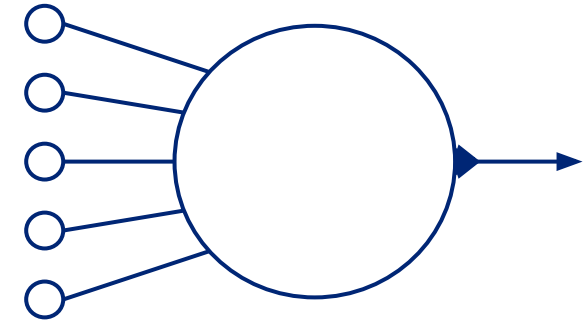
Machine Learning



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Training

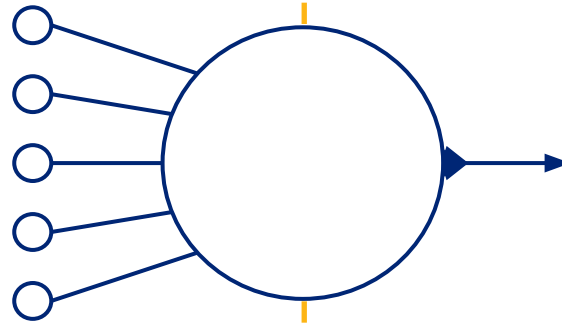


Big Model



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Training



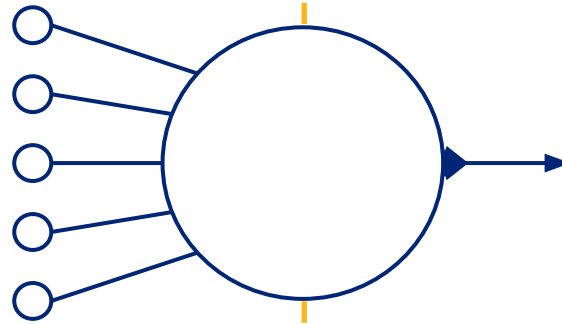
Model



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Training

Inference



Model



Application

When should I use Machine Learning?

What kinds of problems are
Machine Learning Problems?



**Select all options that are a
machine learning problem.**



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Kinds of Problems

Engineering Problem: Can be solved with a direct, specifiable algorithm or a set of hand-written rules.

Machine Learning Problem: For which it is easy to demonstrate or evaluate the solution but difficult to directly implement.

A Human Problem: The problem cannot be well specified and/or human judgement is required.

- Often require *Engineering + ML + Humans* 🤝



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A Machine Learning Problem

A problem for which it is **easy to demonstrate** or **evaluate the solution** but **difficult to directly implement**.

Machine Learning Solution:

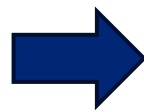
The system **learns the desired behavior** (e.g., prediction, representation, or a policy) through **demonstration** or **experience**.

Exmpl

Determine if a text message is

How do you
define Spam?

Spam is **difficult to define** and
depends on the receiver.



Easier to **demonstrate examples**
and **learn a function to detect**

Is Chatting a Machine Learning Problem?



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Example

(ChatGPT):

Engage a human in a productive conversation

How do you program this?

ELIZA (1966)
Rule based
Conversational
System

```
Welcome to

EEEEEE LL      IIII  ZZZZZZ  AAAAA
EE      LL      II    ZZ     AA  AA
EEEEEE LL      II    ZZZ     AAAAAA
EE      LL      II    ZZ     AA  AA
EEEEEE LLLLLL  IIII ZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

<https://en.wikipedia.org/wiki/ELIZA>

<https://web.njit.edu/~ronkowit/eliza.html>

Entertaining but
it can't do your
homework.

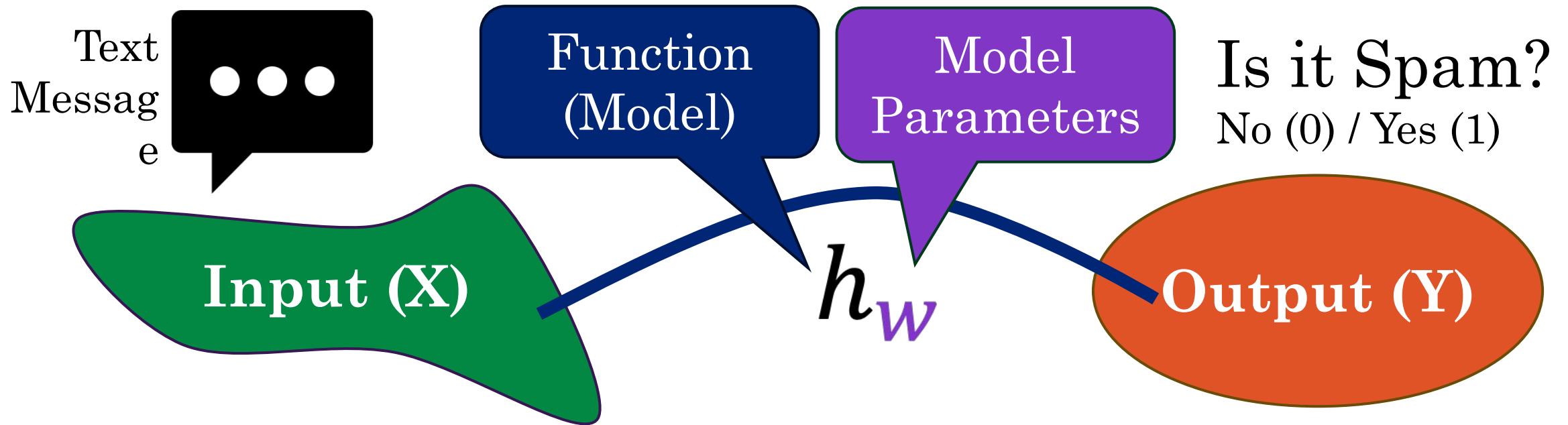
We can demonstrate
good conversations.

We can judge good
conversations.

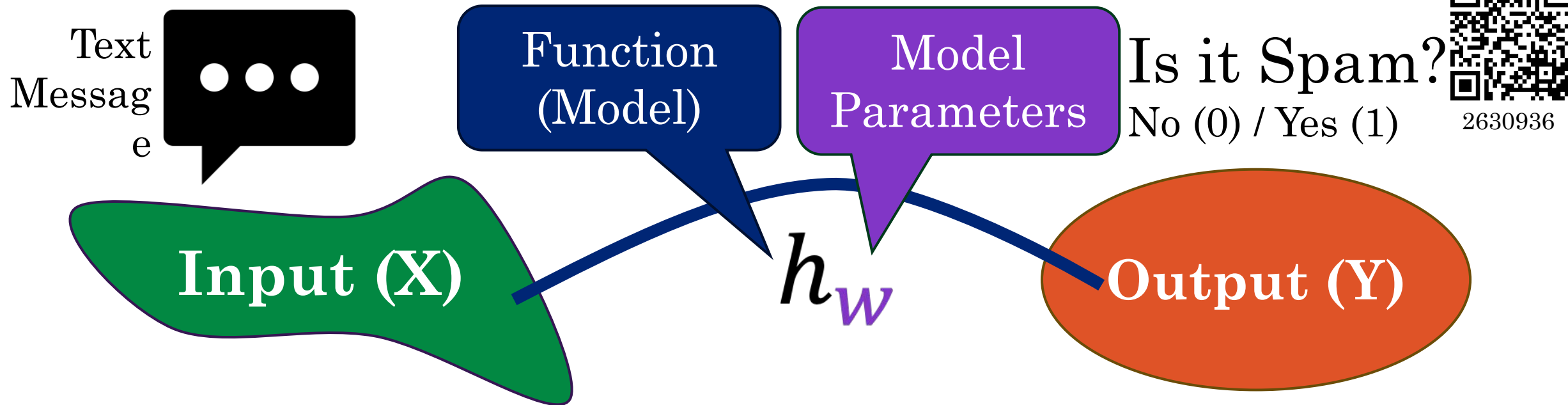
Machine Learning as Learned Function Approximation



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Machine learning becomes the process of “**learning**” the **model parameters** from data or interaction with the world.



Learning Settings

Supervised
(Demonstration)

Unsupervised

Reinforcement
(Reward)

Observation:

$\{(\mathbf{X}, \mathbf{Y})\}$

$\{\mathbf{X}\}$

$\mathbf{X}, \text{reward}(\cdot)$

Supervised Learning



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Trying to learn a relationship between observed $\{(X, Y)\}$ pairs.

Classification

Regression

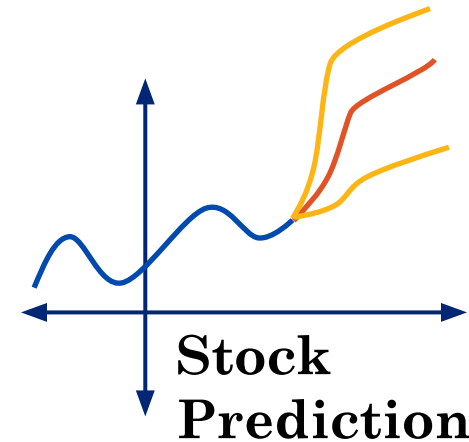


Image Labeling

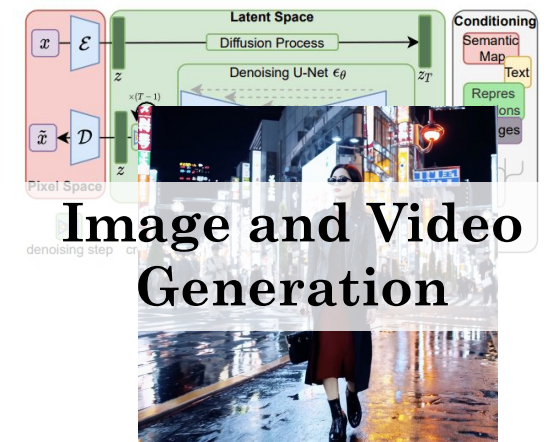
X: Image
Y: {Hot Dog, ...}



X: Prompt
Y: Next Word



X: History
Y: Next Value



X: Prompt+Noise
Y: Pixel Values

Unsupervised Learning

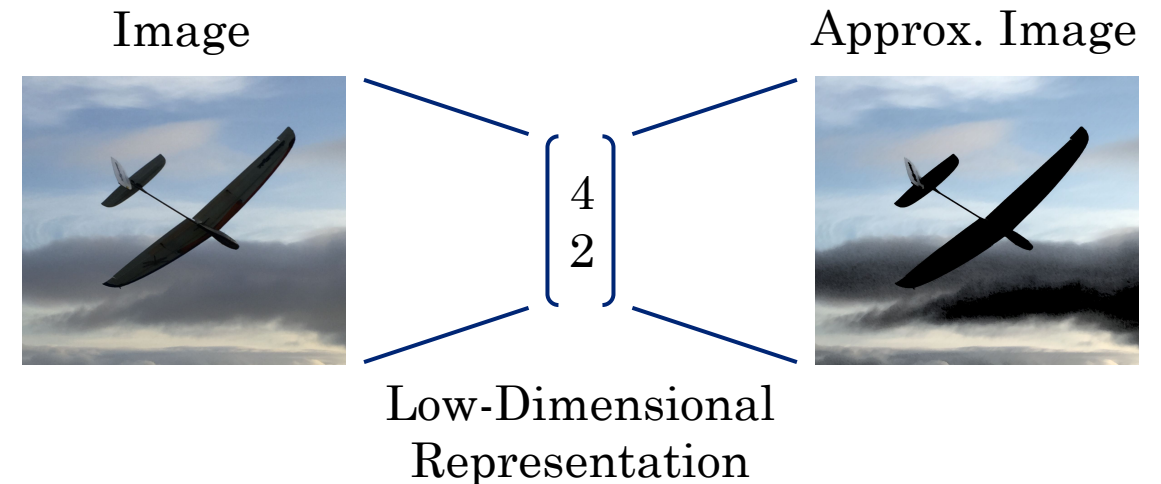


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Trying to **model** the data in the **absence** of **explicit** labels.

**Clustering &
Density Estimation**

**Dimensionality
Reduction**



Used for **visualization** and as a step in **other ML tasks**.

Reinforcement Learning

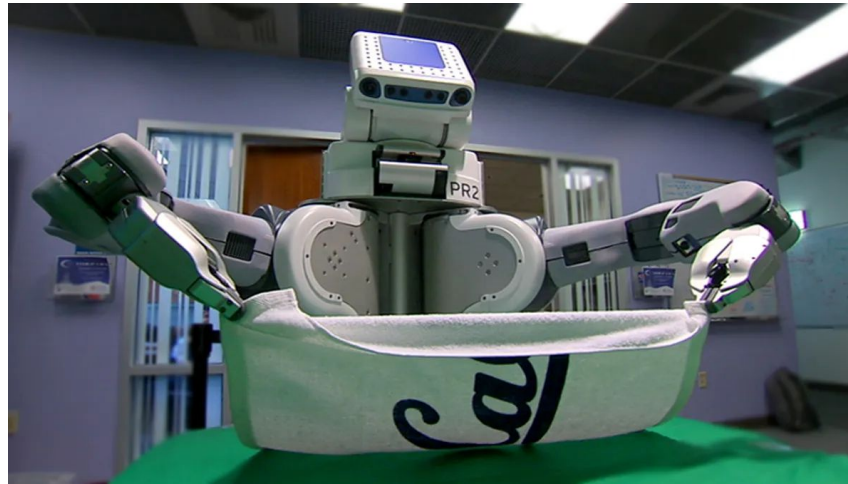


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Learning from **reward signals** often with complex **multi-step** (discrete or continuous) **action** sequences.



Action: next move,
Reward: Win/Lose



Action: change in joint angles
Reward: Fold quality



Action: next token
Reward: answer quality

Not covered in this class but a direct extension of topics in this class.

History of Machine Learning



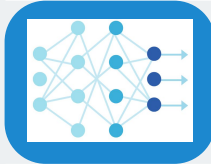
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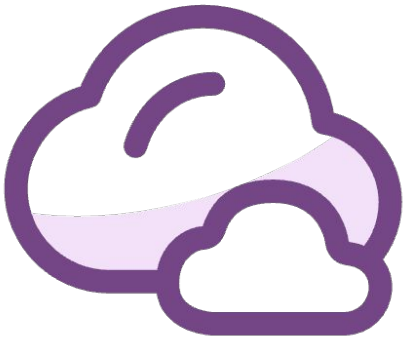
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History of ML



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1950s–60s Early Days	1970s–80s Challenges and Advances	1990s Rise of Statistical ML	2000s Big Data Era	2010s Deep Learning Revolution	Present
Self-learning checkers program (1959) Perceptron (1957)	Decision trees, RL basics, and the rediscovery of NNs	Probabilistic models & statistical learning. Focus on math foundations.	Datasets grew & computation became cheaper Rise of Data Mining & Data Science	Deep learning (2012)	?
					

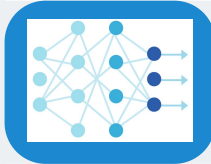


What do you think defines AI today?

History of ML



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Today – GenAI



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What does it mean?

Models that can
generate the data.

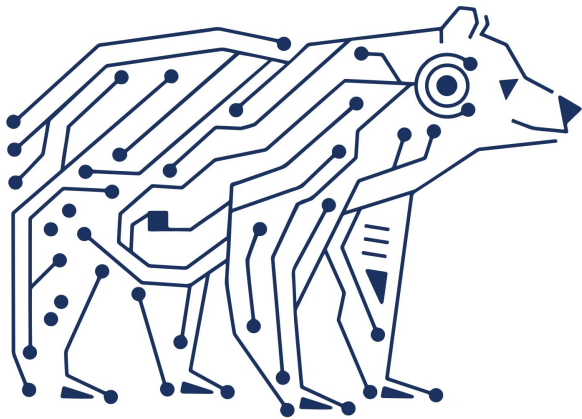
Why is it important?

Unlocking new advanced
general AI abilities.

Will we cover it?

Yes!

UC Berkeley is at the center of the AI revolution:



Discover BAIR, the world's most
advanced academic AI research lab.

[Vicuna](#)
[AdaGrad](#)
[TRPO](#)
[NeRF](#)
[LLM-as-a-Judge](#)
[MMLU](#)
[Spark](#)
[Caffe](#)
[LLM](#)
[Diffusion Models](#)
[Chatbot Arena](#)
[GPT-4 / BEG](#)

sky **rise**lab
UC Berkeley

– **amplab**

Get Involved in
Research!

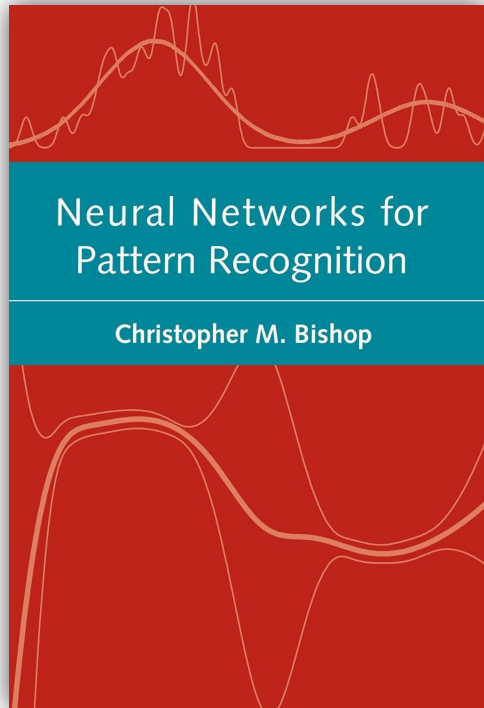
How We teach ML Has *Evolved*



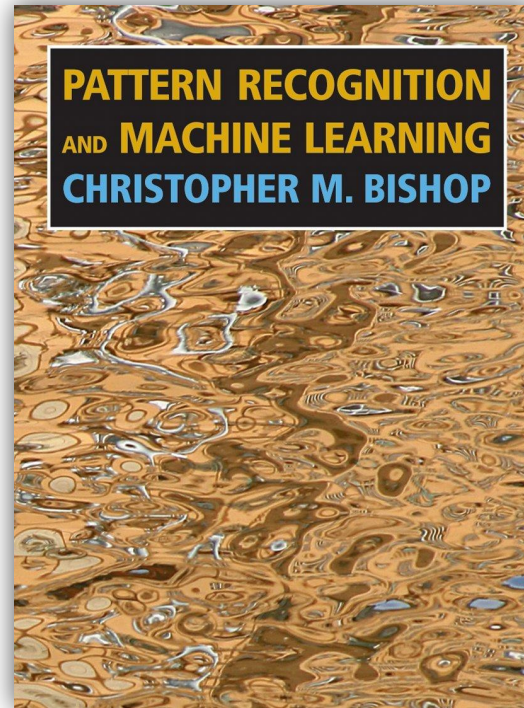
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Probability and Linear Algebra

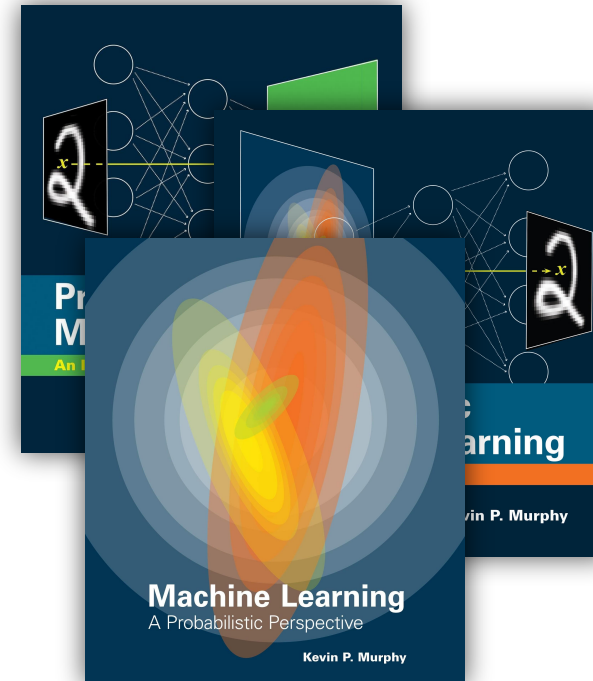
Deep Learning
(NN + Prob + Lin Alg.)



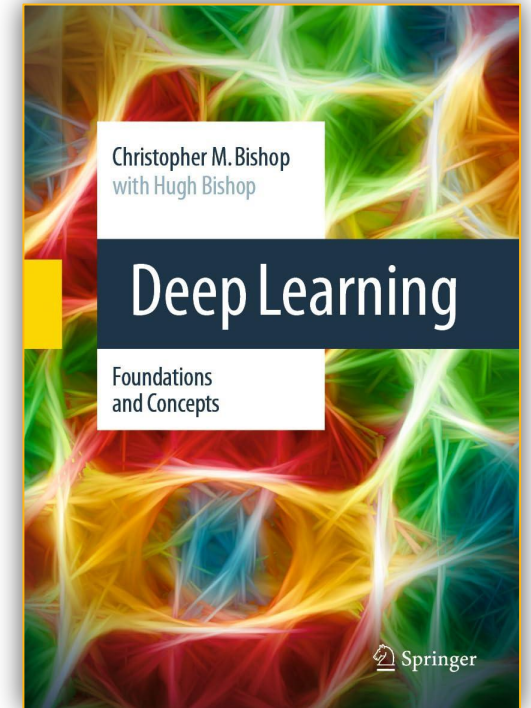
1996



2006



2013-2022



2023

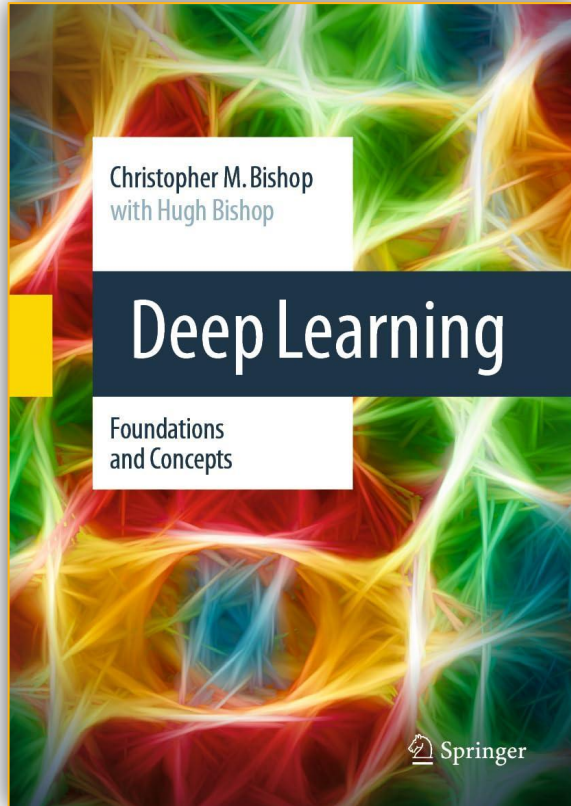


Click on the books to get free PDF versions of all of them!

Teaching CS189 w/ Bishop's Latest Book



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We will follow the books **notation** and **concepts**

- Book should be a helpful resource

Issues with the book (for this class):

- Heavy emphasis on **probabilistic** framing
- Some material is **out-of-scope for this class**
- Doesn't cover **coding related activities**

*Each lecture will have a list of textbook sections that we covered and you're **STRONGLY** encouraged to read the textbook!*



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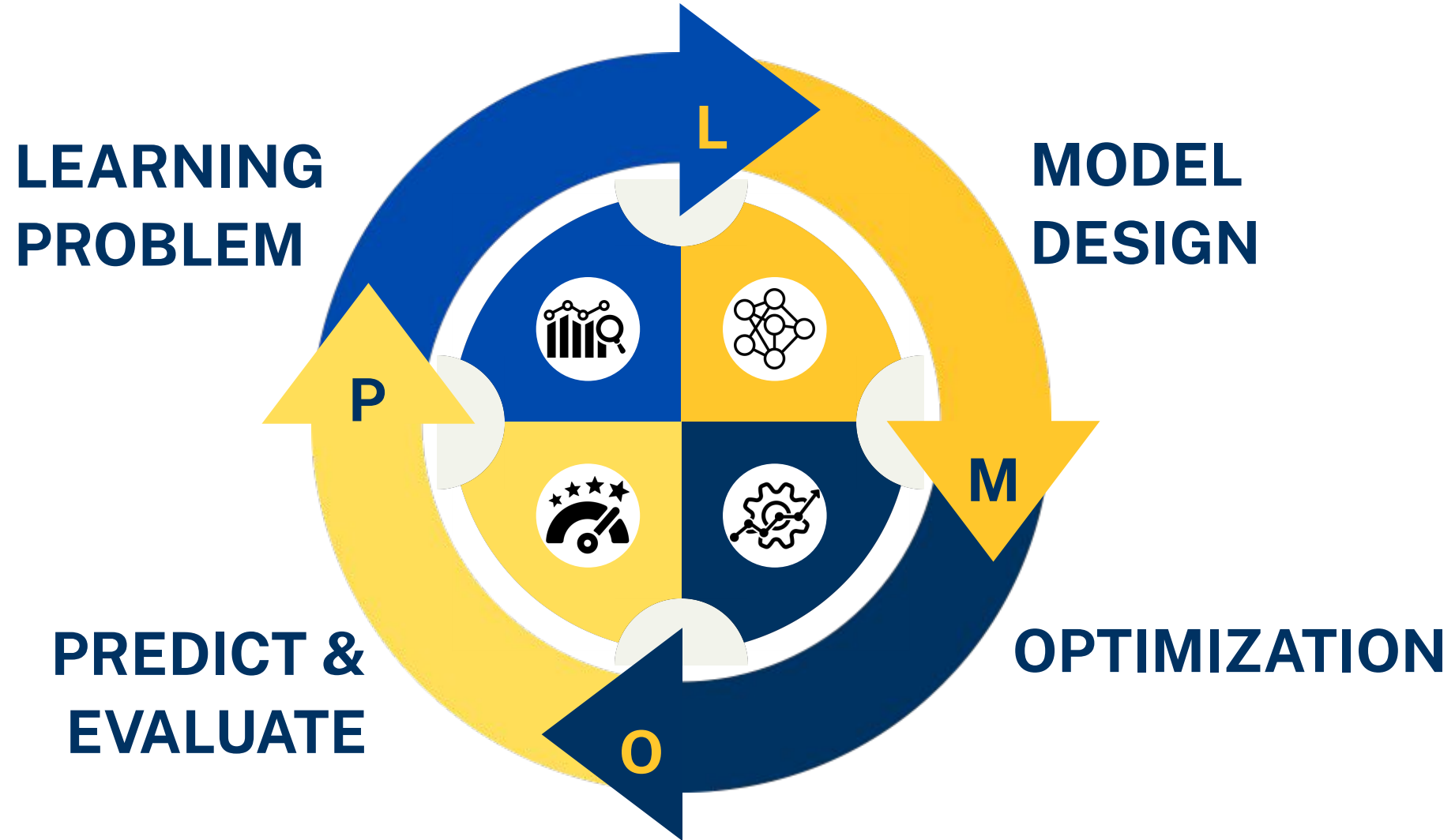
The ML Process (Lifecycle)

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ML Lifecycle



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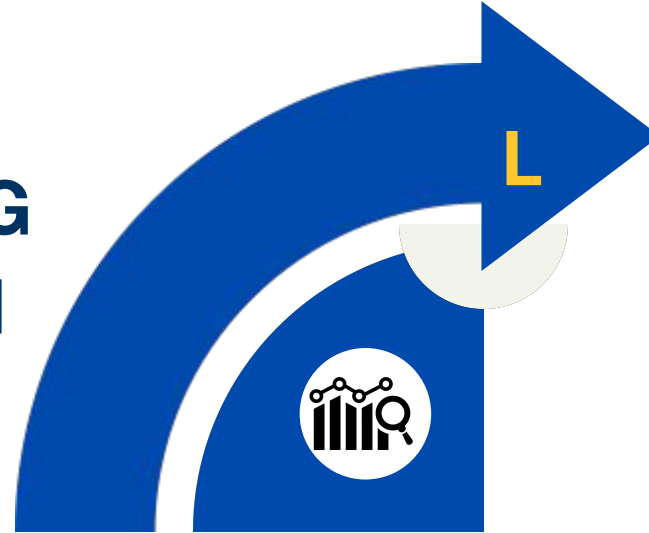
ML Lifecycle

This stage is about framing the real-world question into something a machine learning model can answer.



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LEARNING PROBLEM



- **Target:**
 - What do I want to **predict**? What is the machine learning task?
- **Objective:**
 - How would I **evaluate success**? What loss should I use?
- **Data:**
 - What **data** do I have?
 - Data representation? Feature Engineering?
 - Training/Test split

ML Lifecycle

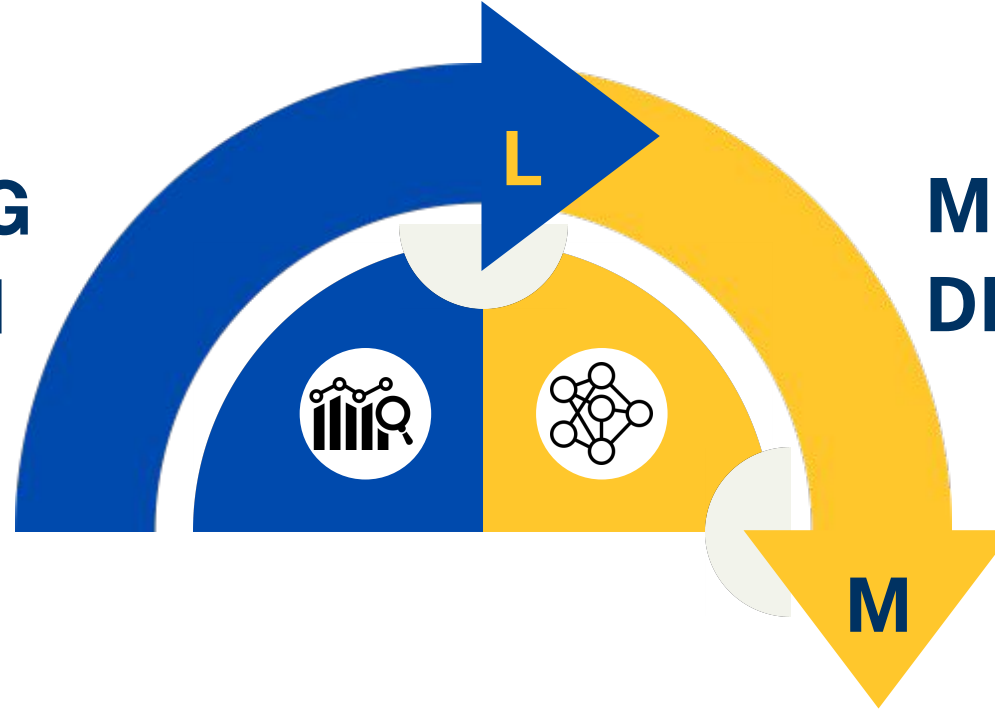
Choose and design an appropriate model.



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LEARNING PROBLEM

- Target
- Objective
- Data



MODEL DESIGN

- Model family/Architecture
- Hypothesis space
- Inductive biases / Assumptions

ML Lifecycle

Adjusting the model's parameters to minimize error using optimization algorithms.



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LEARNING PROBLEM

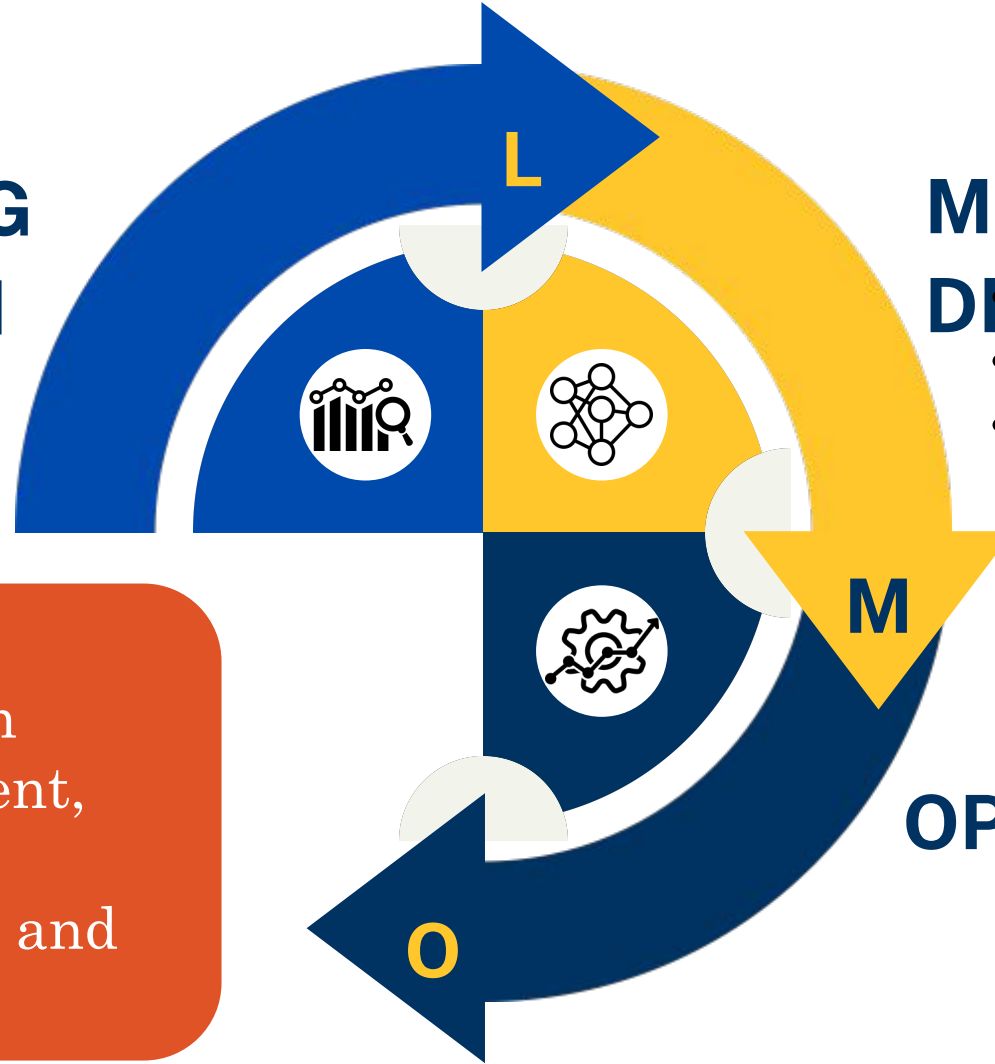
- Target
- Objective
- Data

- Define a loss
- Choose an optimization method (gradient descent, etc.)
- Manage regularization and overfitting

MODEL DESIGN

- Model family/Architecture
- Hypothesis space
- Inductive biases / Assumptions

OPTIMIZATION



ML Lifecycle

Test how well the model performs.



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LEARNING PROBLEM

- Target
- Objective
- Data

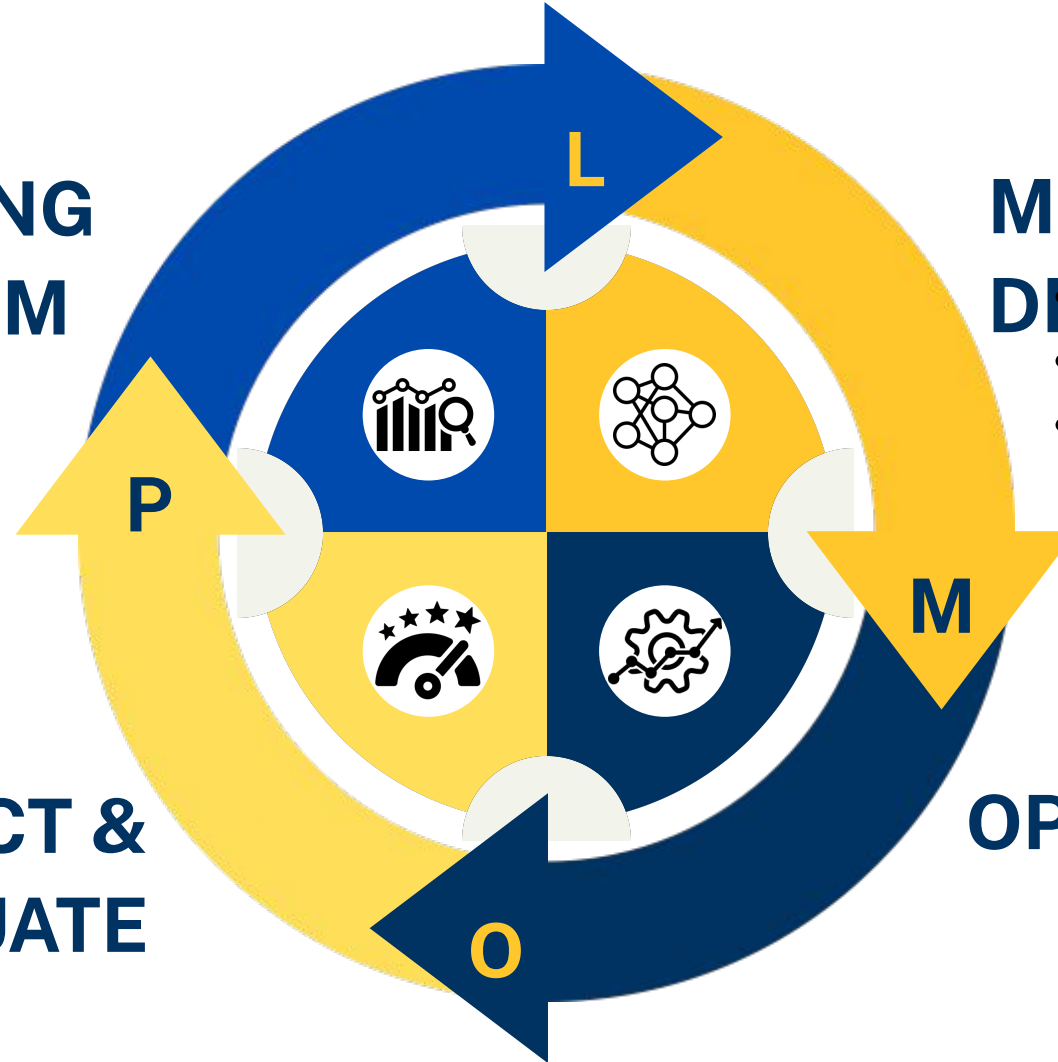
MODEL DESIGN

- Model family/Architecture
- Hypothesis space
- Inductive biases / Assumptions

Evaluate
predictions
based on
evaluation
metrics

PREDICT & EVALUATE

OPTIMIZATION



We are going to ...

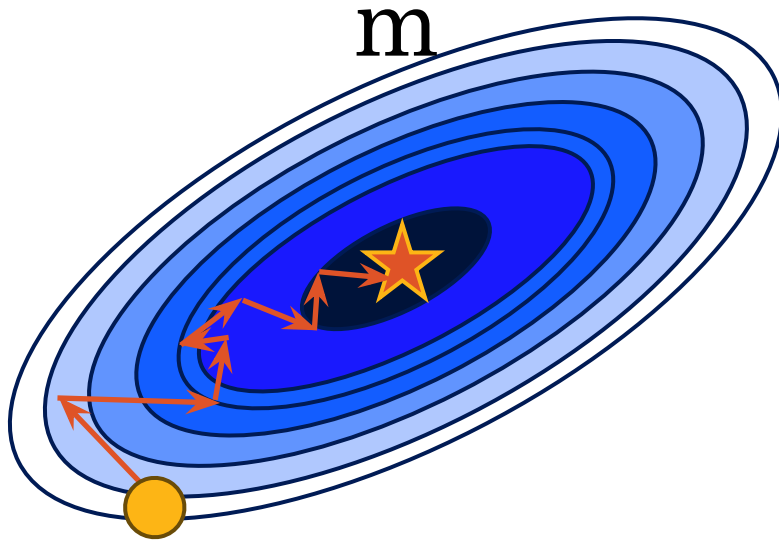
Teach ML “Backwards”

Classic Machine Learning Classes

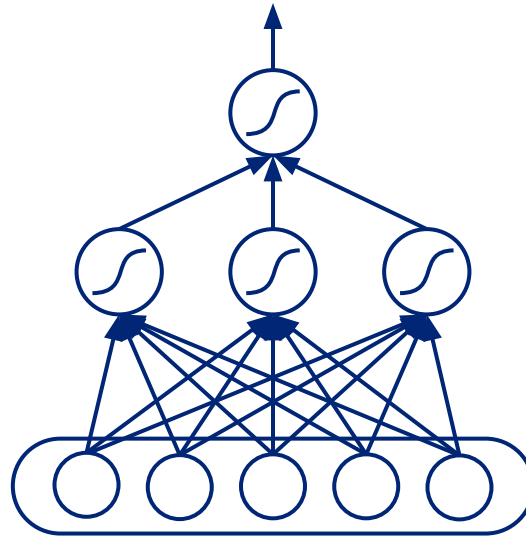


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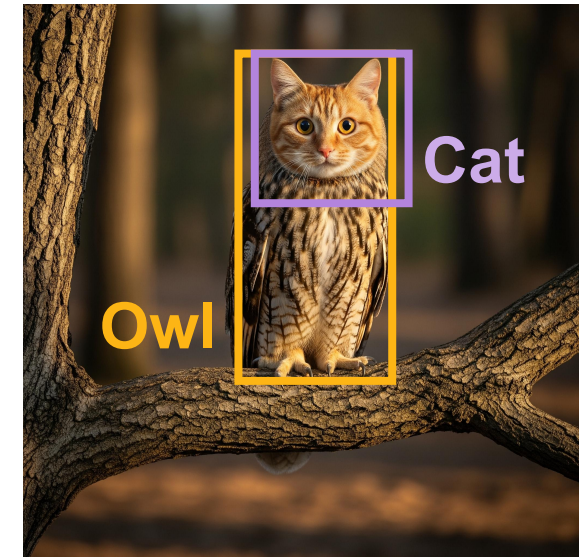
Algorithm



Model



Application



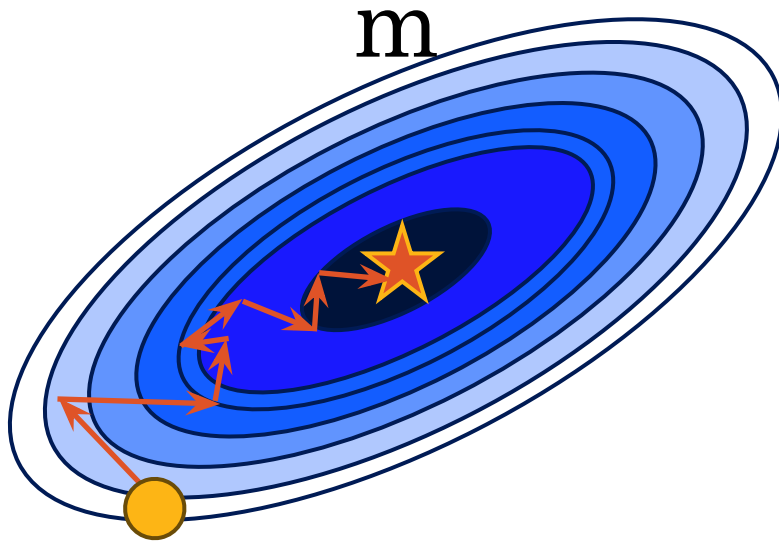
Classic Machine Learning Class

Teaching Machine Learning Differently

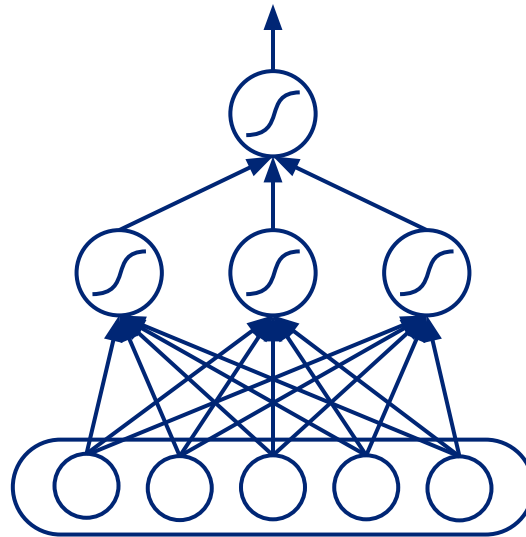


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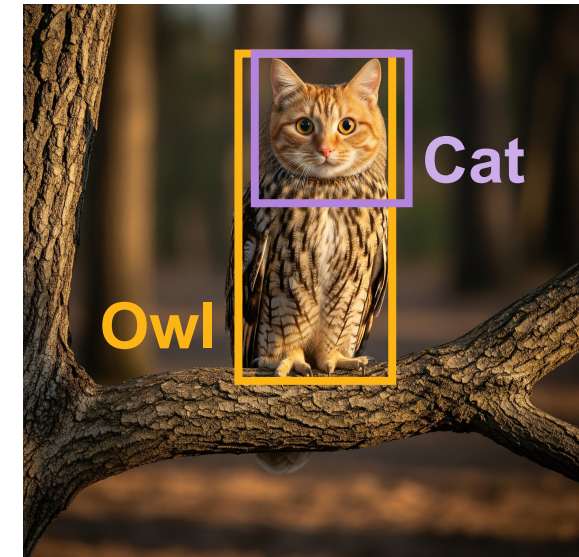
Algorithm



Model



Application



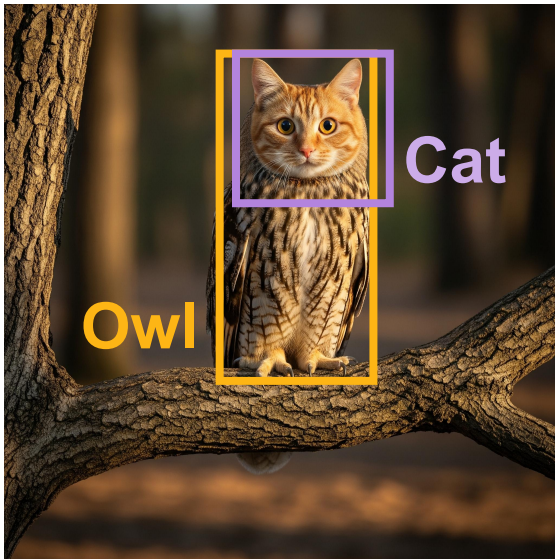
This Machine Learning Class

Greater Focus on Application Framing



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Application



- When should I use machine learning?
- How do I frame a machine learning problem?
- How do I prepare my data?
- How do I train a model?
- How do I evaluate the model?



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Greater Focus on ML Engineering

You will learn to use **tools for ML**:

- **pandas** (<https://pandas.pydata.org/>) for data manipulation.  **pandas**
- **Plotly** (<https://plotly.com/>) and **Matplotlib** (<https://matplotlib.org/>) for data visualization.  **plotly**  **matplotlib**
- **Scikit-learn** (<https://scikit-learn.org/stable/>) for classic machine learning tasks. 
- **PyTorch** (<https://pytorch.org/>) and **Hugging Face** (<https://huggingface.co/>) for neural network development.  **Hugging Face**  **PyTorch**
- **Weights-and-Biases** (<https://wandb.ai/site/>) for experiment management. 

We will work in the **Google Colab environment**, but you will also be able to use your own tools if you prefer.

Logistics

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Course Map at a Glance



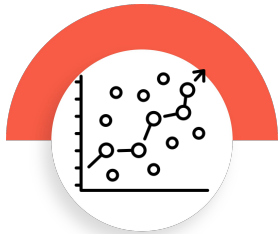
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1-2



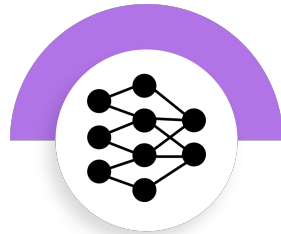
Introduction and
ML Mechanics

3-6



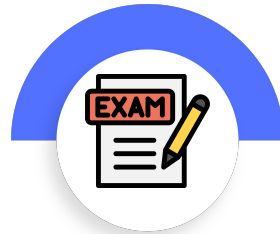
**Supervised
Core**
k-means/EM;
regression →
classification;
GD

7-8



**Neural
Networks**
NN Fundamentals:
backprop
non-linearity
regularization

9



**Midterm
Week**

10-11



**Advanced
Architectures**
CNN
RNN
Transformer
LLM

12-14



**Advanced
Topics**
Generative Models
Autoencoder
Dimensionality
Reduction

15-16



Applications
Guest Lecture
More advanced
applications

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
09:00					OH	 Weekly Beat	
10:00							
11:00		OH			Joey's OH		
12:00			Discussion		HW		
13:00					RELEAS		
14:00		Lecture		Lecture	E		
15:00			Discussion				
16:00	OH		OH				
17:00				Narges' OH			
18:00		Discussion		OH			
19:00							
23:59					HW DUE		

Assessment Cadence



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- **5 Homeworks**, each spans **~3 weeks**, with **two parts** released together:
 - **Part 1 (Warmup)** → introduces tools;
 - **Part 2 (Main)** → deeper application.
- Mix of **written + coding** on Gradescope;
 - Some **public** sanity tests and **hidden** correctness tests.
 - Collaboration = discuss ideas, but **write/code individually**.
 - Open use of GenAI
 - No Vibe Coding/Writing!! You must understand everything you submit!
- **Due 11:59 pm**
 - **No HW drops.**
 - You get **10 total slip days** across the **10 HW deadlines**;
 - **max 4** per HW; slip days **auto-apply** (1 minute late = 1 day).

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Prerequisites



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Prerequisites

- Official Prerequisite:
 - MATH 53 and MATH 54; and COMPSCI 70 (or equivalent).
- What you need:
 - Good programming skills in Python
 - Implement functions and classes
 - Debug programs
 - Probability and statistics
 - Work with conditional probabilities and Bayes rule and compute expectations
 - Familiarity with uniform, Bernoulli, and Gaussian distributions
 - Linear algebra and Calculus
 - Understand eigenvectors and eigenvalues
 - Compute gradients, apply the chain rule

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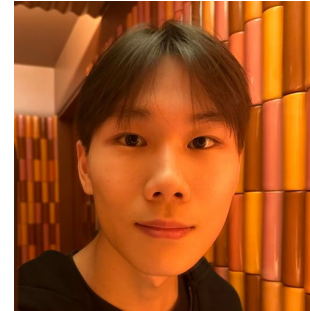
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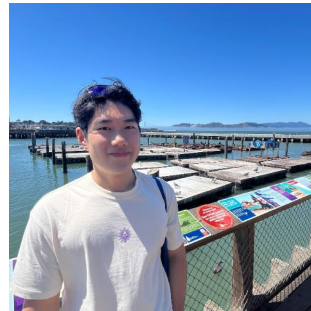
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Course Platforms



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Course Platforms

- Website <https://eecs189.org/fa25/>
- Homework submissions <https://www.gradescope.com/courses/1074470>
- Official programming environment on Google Colaboratory
 - Go signup for a FREE Colab Pro for Education Account
- You may use your own tools (vscode, cursor, jupyterlab, ... vim?)
 - We won't be able to officially support you
- Ed <https://edstem.org/us/courses/81430/>

Course Platforms - Askademia



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Integer-based Extraction: `.iloc`

A different scenario: We want to extract data according to its **numeric position**.

- Example: Grab the 1st, 2nd, and 3rd columns of the `DataFrame`.

`df.iloc[row_integers, column_integers]` `.iloc` stands for integer location

The `.iloc` accessor allows us to specify the **integers** of rows and columns we wish to extract.

- Python convention: The first position has integer index 0.

	0	1	2	3	4	5	Column integers
	Year	Candidate	Party	Popular vote	Result	%	
0	1824	Andrew Jackson	Democratic-Republican	151271	loss	57.210122	
1	1824	John Quincy Adams	Democratic-Republican	113142	win	42.789878	
2	1828	Andrew Jackson	Democratic	642806	win	56.203927	
3	1828	John Quincy Adams	National Republican	500897	loss	43.796073	
4	1832	Andrew Jackson	Democratic	702735	win	54.574789	
...	...	...	...	...	...	...	

All Lectures

0:16 / 2:07

1x

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You at **0:16:**

what is `.iloc`?

Askademia:

`.iloc` is a method in pandas used for selecting data from a `DataFrame` based on **integer-based indexing**. Unlike `.loc`, which uses labels, `.iloc` works with the positions of rows and columns. For example, `df.iloc[0, 1]` retrieves the value in the first row and second column of the `DataFrame`. It's worth noting that slicing with `.iloc` is **exclusive** on the end index, meaning `df.iloc[0:4, 0:4]` selects rows 0 to 3 and columns 0 to 3.

Ask a question...

Send

<https://eecs189.org/fa25/>

Let's Review Syllabus on the Website

Lecture 1

Introduction to Machine Learning

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters: Chapter 1 (Section 1.1)